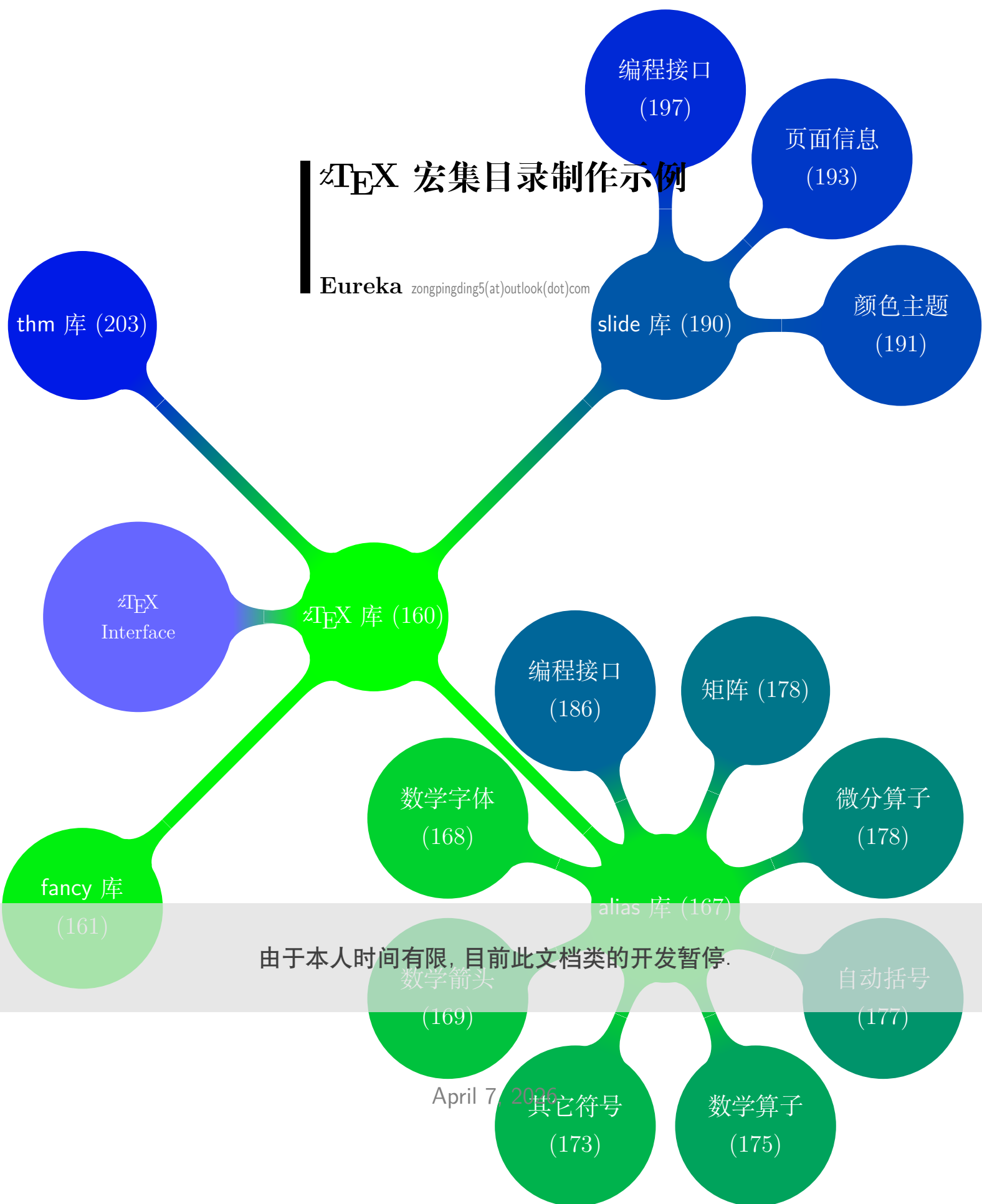


zT_EX 宏集目录制作示例

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1 序言

最后附上一些复杂的目录格式定制示例，涵盖目录样式设置以及目录数据提取，可作为用户深度定制的参考。下面这段代码展示了后续样例会用到的一些命令和依赖宏包：

```

1 % \usepackage{tikz}
2 % \usepackage{tikz-qtrees}
3 % \usepackage[log]{tikz-qtrees-patch}
4 % \usetikzlibrary { mindmap, trees }
5 \definecolor{SeaGreen}{rgb}{0.18, 0.55, 0.34}
6 \def\customtothemecolor{SeaGreen}
7 \ExplSyntaxOn\makeatletter
8 \dim_new:N \linewidthfix
9 \def\tocupdatelinenwidthfix{
10   \dim_set:Nn \linewidthfix {\linewidth-2\fbboxsep}
11 }
12 \def\ornamentcorner#1#2#3{
13   \node[anchor=#1, rotate=#3, outer~sep=0pt, inner~sep=0pt] at (frame.#2)
14     {\pgfornament[width=.8cm, color=\customtothemecolor]{10}};
15 \def\ztochyperpage#1{\exp_args:Nne \hyper@link{link}{page.#1}{#1}}
16 \makeatother\ExplSyntaxOff

```

2 测试数据

本手册在排版目录时用到的测试数据：“data.toc”，“data_II.toc”，“data.ptoc”。data.ptoc 文件仅用于向读者展示 sect 模块中 keyval seq 的存储结构；当“dump-ptable = true”时， $\LaTeX$ 会自动生成该文件。

<pre> \contentsline {section}{1}{AAA-1}{1}{} \contentsline {subsection}{1.1}{BBB-1}{1}{} \contentsline {subsubsection}{1.1.1}{CCC-1}{1}{} \contentsline {subsubsection}{1.1.2}{CCC-2}{1}{} \contentsline {subsubsection}{1.1.3}{CCC-3}{1}{} \contentsline {subsubsection}{1.1.4}{CCC-4}{1}{} \contentsline {subsubsection}{1.1.5}{CCC-5}{1}{} </pre>	data.toc
--	----------

```

\contentsline {subsection}{1.2}{BBB-2}{1}{}
\contentsline {subsubsection}{1.2.1}{CCC-6}{1}{}
\contentsline {subsubsection}{1.2.2}{CCC-7}{1}{}
\contentsline {subsubsection}{1.2.3}{CCC-8}{1}{}
\contentsline {subsubsection}{1.2.4}{CCC-9}{1}{}
\contentsline {subsection}{1.3}{BBB-3}{1}{}

```

```

\contentsline{section}{1}{AAA-1}{1}{}% data_II.toc
\contentsline{subsection}{1}{BBB-1}{1}{}%
\contentsline{subsubsection}{1.1.1}{CCC-1}{1}{}%
\contentsline{subsubsection}{1.1.2}{CCC-2}{2}{}%
\contentsline{subsubsection}{1.1.3}{CCxxxC-3}{3}{}%
\contentsline{subsubsection}{1.1.4}{CCssgda\_shgfeC-4}{100}{}%
\contentsline{subsubsection}{1.1.5}{CsasfCC-5}{123}{}%
\contentsline{subsubsection}{1.1.6}{CasklklCC-x}{999}{}%
\contentsline{subsection}{2}{B BB-2}{1}{}%
\contentsline{subsubsection}{1.2.1}{CCC-6}{1}{}%

```

```

class={section},name=1,title={AAA-1},page=1,anchor={section.1},raw=1,ptoc
\contentsline {section}{1}{AAA-1}{1}{}
class={subsection},name=1.1,title={BBB-1},page=1,anchor={subsection.1.1},
raw={\contentsline {subsection}{1.1}{BBB-1}{1}{}
class={subsubsection},name=1.1.1,title={CCC-1},page=1,
anchor={subsubsection.1.1.1},raw={\contentsline
{subsubsection}{1.1.1}{CCC-1}{1}{}
class={subsubsection},name=1.1.2,title={CCC-2},page=1,
anchor={subsubsection.1.1.2},raw={\contentsline
{subsubsection}{1.1.2}{CCC-2}{1}{}
class={subsubsection},name=1.1.3,title={CCC-3},page=1,
anchor={subsubsection.1.1.3},raw={\contentsline
{subsubsection}{1.1.3}{CCC-3}{1}{}
class={subsubsection},name=1.1.4,title={CCC-4},page=1,
anchor={subsubsection.1.1.4},raw={\contentsline
{subsubsection}{1.1.4}{CCC-4}{1}{}

```

```

class={subsubsection},name={1.1.5},title={CCC-5},page={1}, ✓
anchor={subsubsection.1.1.5},raw={\contentsline ✓
{subsubsection}{1.1.5}{CCC-5}{1}{}}
class={subsection},name={1.2},title={BBB-2},page={1},anchor={subsection.1.2}, ✓
raw={\contentsline {subsection}{1.2}{BBB-2}{1}{}}
class={subsubsection},name={1.2.1},title={CCC-6},page={1}, ✓
anchor={subsubsection.1.2.1},raw={\contentsline ✓
{subsubsection}{1.2.1}{CCC-6}{1}{}}
class={subsubsection},name={1.2.2},title={CCC-7},page={1}, ✓
anchor={subsubsection.1.2.2},raw={\contentsline ✓
{subsubsection}{1.2.2}{CCC-7}{1}{}}
class={subsubsection},name={1.2.3},title={CCC-8},page={1}, ✓
anchor={subsubsection.1.2.3},raw={\contentsline ✓
{subsubsection}{1.2.3}{CCC-8}{1}{}}
class={subsubsection},name={1.2.4},title={CCC-9},page={1}, ✓
anchor={subsubsection.1.2.4},raw={\contentsline ✓
{subsubsection}{1.2.4}{CCC-9}{1}{}}
class={subsection},name={1.3},title={BBB-3},page={1},anchor={subsection.1.3}, ✓
raw={\contentsline {subsection}{1.3}{BBB-3}{1}{}}

```

3 案例:1

一个很简单的案例:

例 1

```

\begingroup
\ztcformat\subsection
{
  format+=\color{teal},
  leader.sep=1pt,
  leader.raise=2.5pt,
  page.width=10pt
}
\ztcgroupinsert{subsection,1,begin}%
{%
  \begin{framed}%
  \pgfornament[width = 2cm,color = teal]{67}%
  \quad\rule[-5em]{.5pt}{10em}%
  \begin{minipage}{.75\linewidth}%
  }
\ztcgroupinsert{subsection,1,end}%
{%
  \end{minipage}%
  \end{framed}%
}
\zlocaltoc{subsection}{4}
\endgroup

```

6.1 font 模块.....	14
------------------	----



6.1.1	字体机制	14
6.1.2	默认字体族	17
6.1.3	新建字体族	17
6.1.4	切换字体	19
6.1.5	LaTeX 接口	20
6.1.6	杂项	24

4 案例:2

下面这个案例清晰地展示了目录中这些 hook 的位置关系, `BG:<int>` 和 `EG:<int>` 代表目录条目自身局部组的开始和结束位置 (这些局部组目前已被移除 – 2025-09-12).

```
\ExplSyntaxOn
\int_new:N \g__test_toc_group_int
\AddToHook{ztoc/tocline/begin}
{
  \fbox{BG:\int_use:N \g__test_toc_group_int}
}
\AddToHook{ztoc/tocline/end}
{
  \fbox{EG:\int_use:N \g__test_toc_group_int}
  \int_gincr:N \g__test_toc_group_int
}
\cs_new:Npn \__debug_ztoc_hook:nn #1#2
{ \ztocgroupinsert{#1}{\fbox{\color{#2}\sffamily #1}} }

% begin/end hooks:
\__debug_ztoc_hook:nn {subsection,1,begin}{red}
\__debug_ztoc_hook:nn {subsection,1,end}{red}
\__debug_ztoc_hook:nn {subsubsection,5,begin}{gray}
\__debug_ztoc_hook:nn {subsubsection,5,end}{gray}
```

例 2


```
% new before/after hooks:
```

```
\__debug_ztoc_hook:nn {subsection,1,before}{blue}
\__debug_ztoc_hook:nn {subsection,1,after}{blue}
\__debug_ztoc_hook:nn {subsubsection,1,before}{teal}
\__debug_ztoc_hook:nn {subsubsection,1,after}{teal}
\__debug_ztoc_hook:nn {subsubsection,3,before}{purple}
\__debug_ztoc_hook:nn {subsubsection,3,after}{purple}
\ExplSyntaxOff
\zlocaltoc{subsection}{4}
```

subsection,1,before	BG:0	
6.1 font 模块		14
EG:0 subsection,1,begin	subsubsection,1,before	BG:1
6.1.1 字体机制		14
EG:1 subsubsection,1,after	BG:2	
6.1.2 默认字体族		17
EG:2 subsubsection,3,before	BG:3	
6.1.3 新建字体族		17
EG:3 subsubsection,3,after	BG:4	
6.1.4 切换字体		19
EG:4	BG:5	
6.1.5 Xe _{La} TeX 接口		20
EG:5 subsubsection,5,begin	subsubsection,5,end	BG:6
6.1.6 杂项		24
EG:6 subsection,1,end	subsection,1,after	

5 案例:3

这个案例相较之上一个案例更加复杂:

```
\begin{group}\makeatletter
\tocupdatelinewidthfix
\def\customtothemecolor{teal}
\ztocformat\subsection
{
  explicit=true,
  code={%
    \noindent\fcolorbox{white}{\customtothemecolor}{\hb@xt@{\linewidthfix
      {\bfseries\large\color{white}\ztocname
        \hskip.75em\relax\ztoctitle\hfill\ztocpage}}}%
  }
}
\ztocformat\subsubsection
{
  format+=\color{\customtothemecolor},
  page.format+=\color{\customtothemecolor},
  name.before=\S\;
}
\ztocgroupinsert{subsection,1,begin}%
{%
  \begin{tcolorbox}[
    enhanced, sharp corners,
    boxrule=.5pt, colback=white,
    left=20pt, colframe=\customtothemecolor,
    overlay={
      \ornamentcorner{north west}{north west}{0}
      \ornamentcorner{north west}{south west}{90}
      \ornamentcorner{north west}{south east}{180}
      \ornamentcorner{north west}{north east}{270}
    }
  ]
```

例 3

```
]
\pgfornament[width = 2.5cm,color = \customtothemecolor]{8}%
\kern-6pt\relax\begin{minipage}{.75\linewidth}%
}
\ztcgroupinsert{subsection,1,end}%
{%
\end{minipage}%
\end{tcolorbox}%
}
\zlocaltoc{subsection}{4}
\makeatother\endgroup
```

6.1 font 模块		14
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	§ 6.1.6 杂项	24

6 案例:4

我们可以使用 `\ztocgroupinsert` 命令复刻 `titletoc` 宏包所提供的样式:

例 4

```

\makeatletter
\def\customssstyle#1{%
  \ztocgroupinsert{subsection,#1,begin}
    {\raggedright\hangindent2.3em\relax\noindent\hskip2.3em\relax 【}
  \ztocgroupinsert{subsection,#1,end}{】 \par}
}
\makeatother
\customssstyle{1} \customssstyle{2}
\customssstyle{3} \customssstyle{4}
{
  \ztocformat\subsubsection{
    explicit=true,
    code={%
      \textit{\ztocname\enskip\ztoctitle\enskip
        (\ztochyperpage{\ztocpage})\ztociflast{}{;}\enskip}}
    }
  \ztocformat\subsection{
    explicit=true,
    code = {%
      \S\;\ztocname\enskip\ztoctitle
      \hskip1em\ztochyperpage{\ztocpage}\par}
    }
  \begin{multicols}{2}
    \zlocaltoc{subsection}{4, 11, 8, 6}
  \end{multicols}
}

```

§ 6.1 font 模块 14

【6.1.1 字体机制 (14); 6.1.2 默认字体族 (17); 6.1.3 新建字体族 (17); 6.1.4 切换字体 (19); 6.1.5 $\LaTeX$ 接口 (20); 6.1.6 杂项 (24)】

§ 6.8 sect 模块 83

【6.8.1 序言 (83); 6.8.2 层级/模板 (84); 6.8.3 章节标题 (85); 6.8.4 章节目录 (91); 6.8.5 使用案例 (105); 6.8.6 编程接口: 初始化 (131); 6.8.7 编程接口: 章节命令 (134); 6.8.8 编程接口: 目

录 (137); 6.8.9 编程接口: 模板 (150); 6.8.10 编程接口: 杂项 (152)】

§ 6.5 thm 模块 41

【6.5.1 用户接口 (42); 6.5.2 定理目录 (48); 6.5.3 高级接口 (51); 6.5.4 环境钩子 (55)】

§ 6.3 page 模块 31

【6.3.1 页面布局 (31); 6.3.2 页眉页脚 (32); 6.3.3 页面水印 (35); 6.3.4 杂项 (37)】

7 案例: 5

这个案例相较之上一个案例更加复杂:

```
\begin{group}\makeatletter
% \setcounter{tocdepth}{3}
\def\customtothemecolor{SeaGreen}
\zotocformat\section
{
  explicit=true,
  code={%
    \noindent\hskip1.5em\fbcolorbox{white}{\customtothemecolor}{\hb@xt@
      \dimexpr\linewidth-2\fbboxsep-1.5em\relax
      {\bfseries\large\color{white}第\zhnumber{\ztocname}节
        \hskip.75em\relax\ztoctitle\hfill\ztocpage}}}%
  }
}
\zotocformat\subsection{
  name.before=\S\;, name.width=2.7em,
  format+=\color{\customtothemecolor},
  page.format+=\color{\customtothemecolor},
  leader.sep=2pt, page.width=1em,
  hyper.title=true,
  leader.content=\textcolor{\customtothemecolor}{.},
}
\zotocgroupinsert{section,1,before}%
{%
  \begin{tcolorbox}[
    enhanced, sharp corners,
    boxrule=.5pt, colback=white,
    left=20pt, colframe=\customtothemecolor,
    overlay={
      \ornamentcorner{north west}{north west}{0}
      \ornamentcorner{north west}{south west}{90}
    }
  ]
}
```

例 5

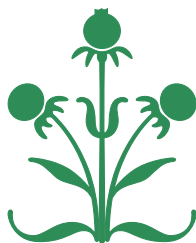
```

\ornamentcorner{north west}{south east}{180}
\ornamentcorner{north west}{north east}{270}
}
]
\pgfornament[width = 2.5cm,color = \customtocthemecolor]{10}%
\begin{minipage}{.77\linewidth}%
}
\ztcgroupinsert{section,1,begin}{%
% \setlength{\columnsep}{0em}
\begin{multicols}{2}}
\ztcgroupinsert{section,1,after}{%
{%
\end{multicols}%
\end{minipage}%
\end{tcolorbox}%
}
\zlocaltoc{section}{6}
\makeatother\endgroup

```

第六节 $\LaTeX$ 模块

13



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8 案例: 6

这个案例复刻了 etoc 中的目录样式 (参见 etoc 宏包第 39 节):

```

1 \ztcgroupinsert{section,1,before} 1
2   {\begin{longtable}{|l|l|r|}} 2
3 \ztcgroupinsert{section,4,after} 3
4   {\ \hline\end{longtable}} 4
5 { 5
6   \ztcformat\section{ 6
7     explicit=true, 7
8     code = {% 8
9       \ztciffirst{\kill\hline\multicolumn{3}{|c|}{\large\bfseries Table of 9
Contents}\ \hline}{\ \hline}%
10       \multicolumn{3}{|c|}{\bfseries 第\;\ztocname\;节\enskip\ztoctitle}% 10
11     } 11
12   } 12
13   \ztcformat\subsection{ 13
14     explicit=true, 14
15     code = {\ \hline 15
16       \textbf{\S\ztocname} & \ztocname\enskip\ztoctitle & \ztohyperpage{ 16
\ztocpage}
17     } 17
18   } 18
19   \ztcformat\subsubsection{ 19
20     explicit=true, 20
21     code = {\ \ 21
22       & \ztocname\enskip\ztoctitle & \ztohyperpage{\ztocpage} 22
23     } 23
24   } 24
25   \zlocaltoc{section}{1, 2, 7, 9} 25
26 } 26

```

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第 9 节 TODO		

9 案例: 7

这个案例也是受到 etoc 宏包的启发 – 将目录展示为思维导图的形式:

```

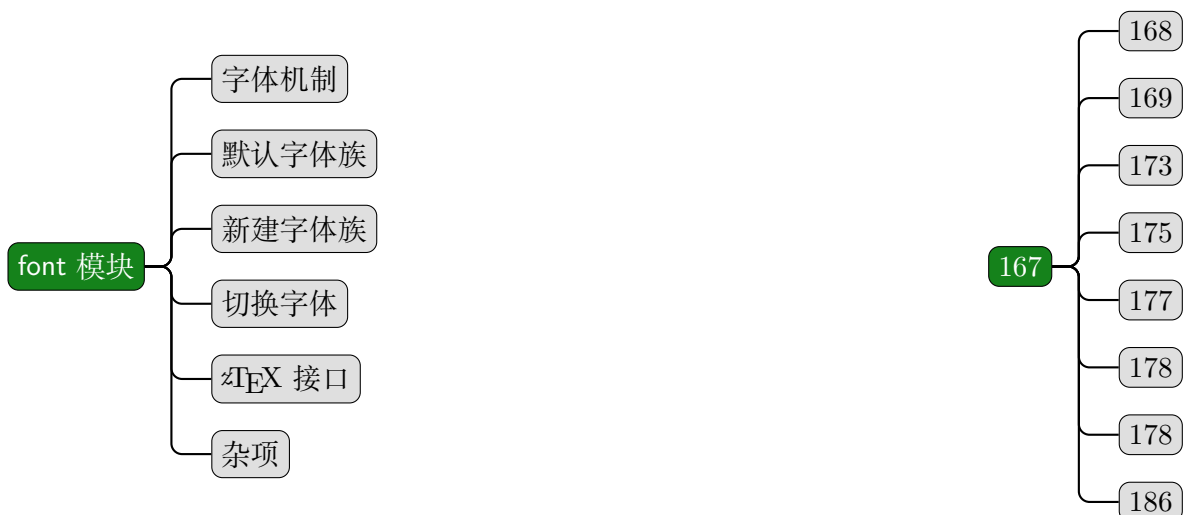
1 \definecolor{Green}{HTML}{15821b} 1
2 \tikzset{ 2
3   rootKey/.style={draw, rectangle, rounded corners, fill=Green, text=white}, 3
4 } 4
5 \ExplSyntaxOn 5
6 \cs_new:Npn \__toc_tree:nn #1#2 6
7 { 7
8   \quark_if_no_value:nF {#1} 8
9   { 9
10     \Tree[.\node[rootKey]{#1};~#2 ] 10
11   } 11
12 } 12
13 \cs_generate_variant:Nn \__toc_tree:nn {oe} 13
14 \def\wrapper#1[. {\exp_not:n {#1}}~] 14
15 \cs_new:Npn \ztoc_typeset_toc tree:nnn #1#2#3 15
16 { 16
17   \ztoc_table_filter_key_byclass:nnnNN {#1}{#2}{#3} 17
18   \g_ztoc_keyvaltoc_seq \g_tmpb_seq 18
19   \seq_gpop_left:NN \g_tmpb_seq \secRoot 19
20   \__toc_tree:oe { \secRoot } 20
21   { \seq_map_function:NN \g_tmpb_seq \wrapper } 21
22 } 22
23 \def\TocTree#1#2#3 23
24 { 24
25   \exp_args:Nne \ztoc_typeset_toc tree:nnn 25
26   {#1}{#2}{#3} 26
27 } 27
28 \ExplSyntaxOff 28
29 29
30 \noindent\begin{tikzpicture}[ 30

```

```

31     grow'=right, 31
32     level distance=50pt, 32
33     sibling distance=10pt, 33
34     level 1/.style={level distance=25pt}, 34
35     every tree node/.style={anchor=west, draw, rectangle, rounded corners, 35
fill=gray!25},
36     every level 0 node/.style={anchor=east}, 36
37     edge from parent/.style={ 37
38         draw, thick, rounded corners, 38
39         edge from parent path={ 39
40             (\tikzparentnode.east) -- +(10pt, 0pt) 40
41             |- (\tikzchildnode.west) 41
42         } 42
43     }, 43
44 ] 44
45 \TocTree{subsection}{4}{title} 45
46 \begin{scope}[xshift=12cm] 46
47     \TocTree{subsection} 47
48     {\ztocindex{subsection}{\pkg{alias} 库}} 48
49     {page} 49
50 \end{scope} 50
51 \end{tikzpicture} 51

```



10 案例: 8

这个案例相较之上一个案例更加复杂, 它展示了 `\ztoc_group_parser:NNNnnnnn` 命令中 `\step code` 的使用方法:

```

1 % \usetikzlibrary { mindmap, trees } 1
2 \ExplSyntaxOn 2
3 \int_new:N \g__mindmap_color_int 3
4 \seq_new:N \g__mindmap_data_seq 4
5 \seq_new:N \g__mindmap_keyvaldata_seq 5
6 \tl_new:N \g__mindmap_filter_data_tl 6
7 \def\toclinecmd#1#2#3#4 7
8 { 8
9   % override hyperlink color. 9
10   {\ztexlink{#1.#2}{\textcolor{white}{#3(#4)}}} 10
11 } 11
12 \cs_new:Npn \__ztoc_gen_mindmap_data:nn #1#2 12
13 { 13
14   \tl_gclear:N \g_tmpa_tl 14
15   \ztoc_table_filter_byclass:nnNN 15
16   { #1 }{ #2 } 16
17   \g_ztoc_keyvaltoc_seq 17
18   \g__mindmap_data_seq 18
19   \ztoc_generate_keyvaltable_seq:NN 19
20   \g__mindmap_data_seq 20
21   \g__mindmap_keyvaldata_seq 21
22   \ztoc_group_parser:NNNnnnnn \g__mindmap_keyvaldata_seq 22
23   \toclinecmd \g_tmpa_tl 23
24   { 24
25     child [ concept~color = blue! 25
26       \int_eval:n{\g__mindmap_color_int*6}!green 26
27     ] \bgroup node[concept] 27
28   } 28
29   {}{}{\egroup}{\int_gincr:N \g__mindmap_color_int} 29

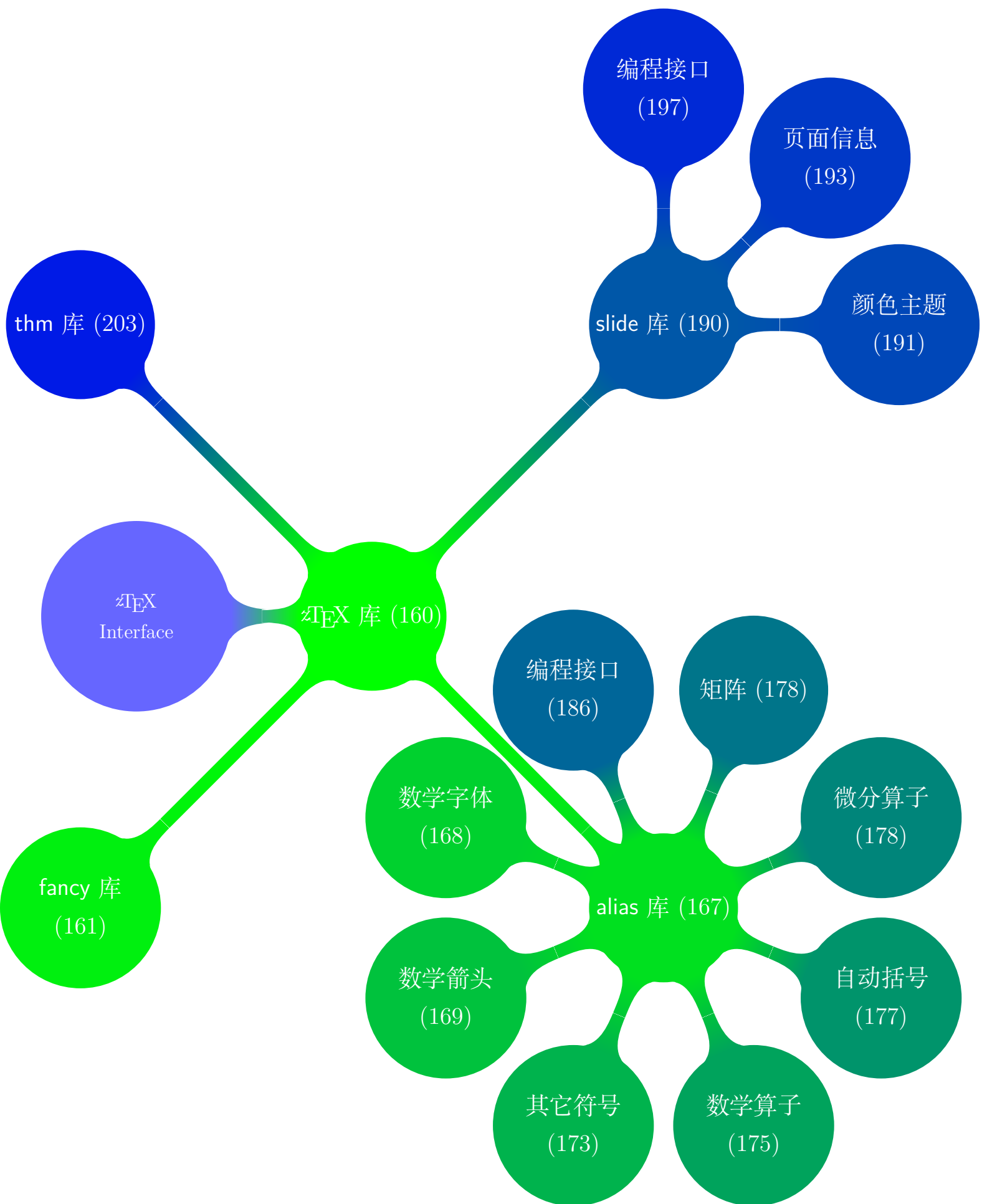
```

```

30 % replace '\bgroup' and '\egroup' by '{' and '}'.
31 \str_clear:N \l_tmpa_str
32 \exp_args:NNo \str_set:Nn \l_tmpa_str { \g_tmpa_tl }
33 \exp_args:NNne \str_replace_all:Nnn \l_tmpa_str
34 { \bgroup }{ \char_generate:nn {123}{12} }
35 \exp_args:NNne \str_replace_all:Nnn \l_tmpa_str
36 { \egroup }{ \char_generate:nn {125}{12} }
37 \tl_gset_rescan:Nno \g__mindmap_filter_data_tl
38 { \cctab_select:N \c_document_cctab }
39 { \l_tmpa_str }
40 }
41 \cs_new:Npn \__tkz_mind_map:n #1
42 {
43   \tl_if_empty:nF {#1}
44   {
45     \node[concept]
46       {\color{white}\ztext{}~Interface}
47       #1;
48   }
49 }
50 \newcommand{\mindmap}[2]
51 {
52   \__ztoc_gen_mindmap_data:nn
53     { #1 }{ #2 }
54   \exp_args:No \__tkz_mind_map:n
55     { \g__mindmap_filter_data_tl }
56 }
57 \ExplSyntaxOff
58 \begin{tikzpicture}[
59   mindmap, grow cyclic,
60   text width=2cm, align=flush center,
61   nodes={concept}, concept color=blue!60,
62   root concept/.append style={

```

```
63      text width=4cm, font=\Large 63
64  }, 64
65  level 1 concept/.append style={ 65
66      font=\normalsize, color=white 66
67  }, 67
68  level 1/.append style={ 68
69      level distance=5cm, sibling angle=45, 69
70      text width=3cm, font=\Large 70
71  }, 71
72  level 2/.append style={ 72
73      level distance=9cm, sibling angle=90, 73
74      text width=3cm, font=\Large 74
75  }, 75
76  level 3/.append style={ 76
77      level distance=5cm, sibling angle=45, 77
78      text width=3cm, font=\Large 78
79  }, 79
80  ] 80
81  \mindmap{section}{7} 81
82  \end{tikzpicture} 82
```



11 案例: 9

下面这个案例展示了 `\ztociffirst`, `\ztociflast` 以及 `\ztocdepth` 等命令的使用方法:

```

1  \ExplSyntaxOn
2  \newcommand{\nbraket}
3    { \makebox[0pt]{\rule[-7.5pt]{.75pt}{19pt}} }
4  \newcommand{\ubraket}
5    {
6      \makebox[0pt]{
7        \rule[-7pt]{.75pt}{10pt}
8      }
9      \makebox[0pt][l]{
10        \kern-.375pt\raise3pt\hbox{\rule{3pt}{.75pt}}
11      }
12    }
13 \newcommand{\bbraket}
14 {
15   \makebox[0pt]{ \rule[3pt]{.75pt}{10pt} }
16   \makebox[0pt][l]
17     { \kern-.375pt\raise3pt
18       \hbox{\rule{3pt}{.75pt}} }
19 }
20 \newcommand{\showbar}
21 {
22   \prg_replicate:nn {\ztocdepth - 2}
23     { \kern20pt \nbraket }
24   \kern20pt \ztociffirst { \ubraket }
25     { \ztociflast{\bbraket}{\nbraket} }
26 }
27 \ztocformat{\section, \subsection, \subsubsection}
28 {
29   explicit=true,
30   code = {

```



```
31      \noindent 31
32      \prg_replicate:nn {\ztocdepth-1} 32
33      { 33
34          \hskip 20pt\relax 34
35      } 35
36      \llap{\smash{\showbar}} 36
37      \kern10pt 37
38      \ztoctitle\par 38
39  } 39
40  } 40
41  \ExplSyntaxOff 41
42  { 42
43      \ztocformat\subsection{ignore.name={10.}} 43
44      \ztocformat\subsubsection{ignore.name={10.}} 44
45      \multicolcontents{2} 45
46  } 46
```

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12 案例: 10

下面这个案例相较之上一个案例更加复杂, 但它展示了 `\ztoc_group_parser:NNNnnnnn` 命令的强大和普适性:

```

1 % \usepackage{tikz} 1
2 % \usepackage{tikz-qtrees} 2
3 % \usepackage{tikz-qtrees-patch} 3
4 \definecolor{Red}{HTML}{e32b2d} 4
5 \definecolor{Green}{HTML}{15821b} 5
6 \definecolor{Blue}{HTML}{0984e3} 6
7 \tikzset{ 7
8   section/.style={draw, rectangle, rounded corners, fill=Red}, 8
9   subsection/.style={draw, rectangle, rounded corners, fill=Blue, text=white}, 9
10  subsubsection/.style={draw, rectangle, rounded corners, fill=Green, 10
    text=white},
11 } 11
12 \ExplSyntaxOn 12
13 % \def\toclinecmd#1#2#3#4{{#1:#2:#3:#4}~} 13
14 \def\toclinecmd#1#2#3#4{{\gparserclass(\gparserindex):#2:#3:#4}~} 14
15 \ztoc_group_parser:NNNnnnnn \g_ztoc_keyvaltoc_seq 15
16 \toclinecmd \g_tmpa_tl {[.]{}}{~}]{} 16
17 \def\TocTree 17
18 { 18
19   \exp_args:Nno \__toc_tree:nn 19
20   {\ztex{}~接口文档} 20
21   {\g_tmpa_tl} 21
22 } 22
23 \cs_new:Npn \__toc_tree:nn #1#2 23
24 { 24
25   \quark_if_no_value:nF {#1} 25
26   { 26
27     \Tree[.\node[section]{#1};~#2 ] 27
28   } 28

```

```

29     }
30     \ExplSyntaxOff
31
32     \thispagestyle{empty}
33     \begin{tikzpicture}[
34         grow'=right, level distance=1cm,
35         sibling distance=10pt,
36         every tree node/.style = {
37             anchor=west, outer sep=0pt, draw, rectangle,
38             rounded corners, fill=gray!25
39         },
40         every level 1 node/.style = {
41             subsection
42         },
43         every level 2 node/.style = {
44             subsubsection
45         },
46         edge from parent/.style={
47             draw, thick, rounded corners,
48             edge from parent path={
49                 (\tikzparentnode.east) -- +(.5cm, 0pt)
50                 |- (\tikzchildnode.west)
51             },
52         },
53     ]
54     \TocTree
55     \end{tikzpicture}

```

